

TopoMedKD: Shape- and Topology-Aware Distillation of Zero-Shot MedSAM for Compact Medical Image Segmentation

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Abstract

- MedSAM: a strong but heavy, prompt-dependent segmentation prior.
- **TopoMedKD** distills a **frozen MedSAM teacher** (GT box, *training only*) into a **compact, prompt-free student** (inference).
- Combines logit, prototype-guided contrastive, and shape/topology-aware distillation; supervision stays dominant.
- Consistently improves lightweight students on KiTS19 kidney-tumour CT.

The problem

- **MedSAM** is powerful but **heavy and prompt-dependent**.
- **Compact students** are efficient but may **miss small tumours** (foreground often <1% of a slice).

Goal: compact, prompt-free student with structural knowledge from MedSAM

Method

Frozen MedSAM (GT box, *training only*) → compact, prompt-free student.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \alpha \mathcal{L}_{\text{LD}} + \beta \mathcal{L}_{\text{PGCD}} + \gamma \mathcal{L}_{\text{TGD}}$$

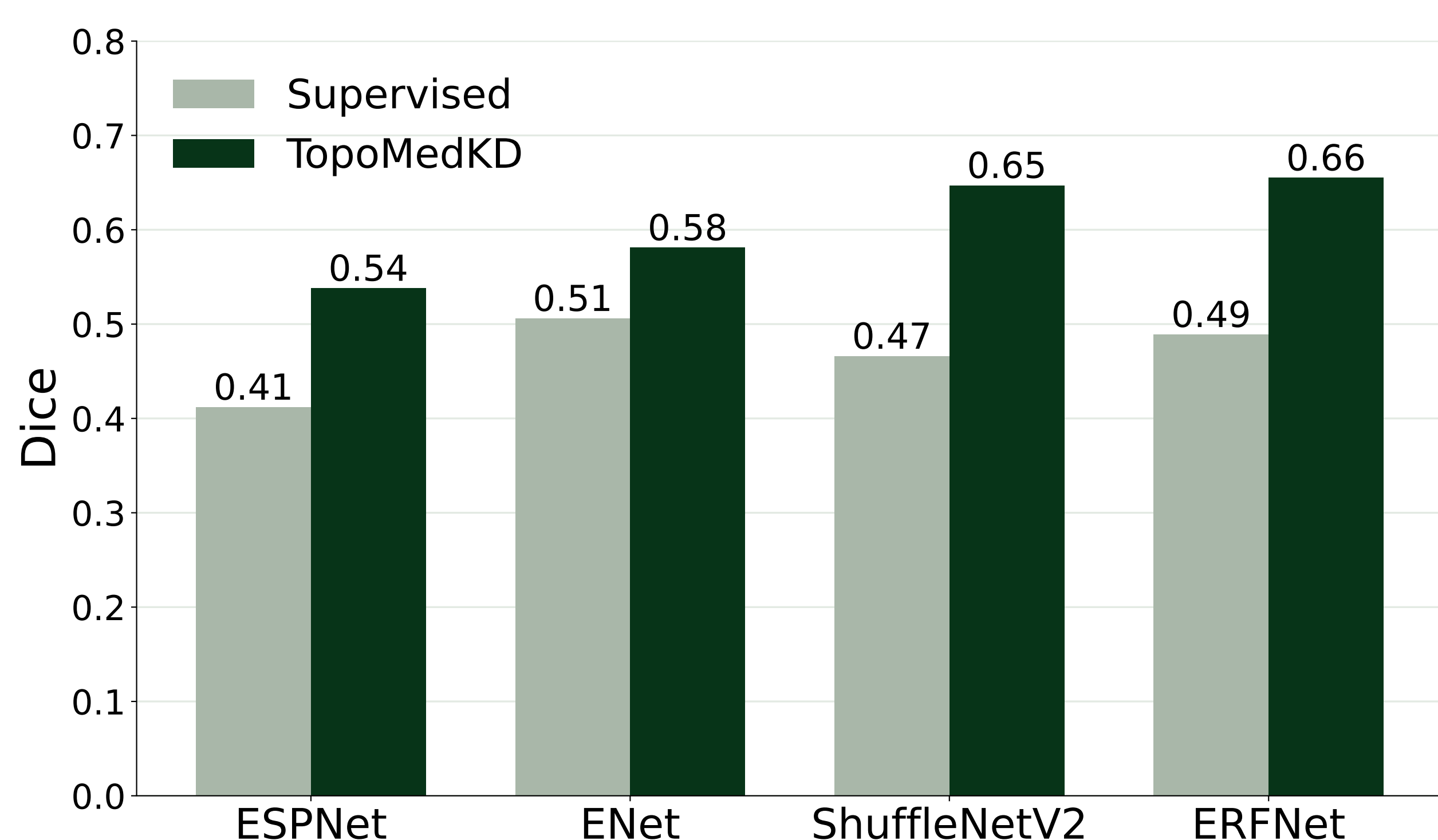
Small weights (α, β, γ) keep supervision dominant.

- **LD:** transfers teacher confidence.
- **PGCD:** aligns class prototypes.
- **TGD:** encourages shape and connectivity consistency.

Results — KiTS19

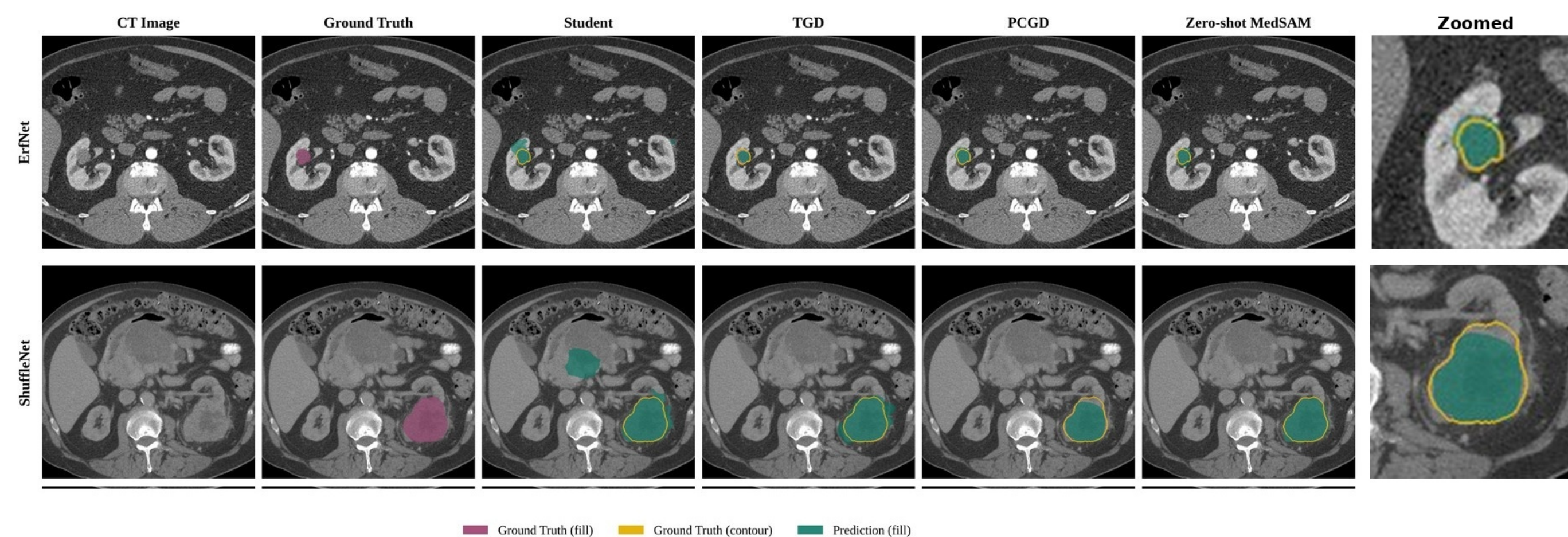
Every student improves over its supervised baseline (zero-shot teacher, never trained on KiTS19). HD95 = surface error (px). Students span two orders of magnitude in parameter count (0.18M–2.06M), yet each one gains from the same frozen teacher without any teacher-side retraining.

Student	Method	Dice ↑	VOE ↓	HD95 ↓ px
ESPNNet	Sup	0.412	0.592	124.1
	Ours	0.538	0.307	80.3
ENet	Sup	0.506	0.401	66.4
	Ours	0.581	0.259	60.4
ShuffleNetV2	Sup	0.466	0.659	155.7
	Ours	0.647	0.375	80.1
ERFNet	Sup	0.489	0.630	120.1
	Ours	0.655	0.281	61.4



ENet, which starts from the strongest supervised baseline (0.506 Dice), gains the least (+0.08); ShuffleNetV2 and ERFNet, with weaker baselines, gain the most (+0.18 and +0.17), consistent with distillation acting as a regulariser for under-parameterised models.

Qualitative



Columns: CT image · ground truth · Student (supervised baseline) · TGD · PGCD (distilled variants) · zero-shot MedSAM (frozen teacher), plus a **zoomed** crop of the PGCD prediction. ■ GT fill ■ GT contour ■ Prediction.

In both rows, the distilled predictions follow the GT contour rather than bleeding into the surrounding healthy kidney tissue (renal parenchyma), and remain a *single connected region*, the fragmentation failure mode TGD is designed to suppress. The zoomed crop makes this concrete: for the large, irregular tumour in the ShuffleNetV2 row, the teal prediction tracks the yellow GT boundary closely rather than only approximately overlapping it; for the smaller tumour in the ERFNet row, the predicted region still separates cleanly from the neighbouring kidney tissue.

Ablation — ShuffleNetV2

The three signals are complementary, gains stack rather than saturate.

LD	PGCD	TGD	Dice ↑	HD95 ↓ px
—	—	—	0.466	155.7
✓	—	—	0.566	97.2
✓	—	✓	0.606	95.3
✓	✓	—	0.615	90.9
✓	✓	✓	0.647	80.1

LD gives the largest gain; PGCD and TGD add complementary improvements.

Zero-shot vs. fine-tuned teacher (ENet)

Teacher	Dice ↑	HD95 ↓ px	Teacher cost
Zero-shot (0.799)	0.581	60.4	none
Fine-tuned (0.762)	0.546	43.7	encoder adaptation

Fine-tuning sharpens boundaries (HD95 ↓) but narrows the posterior/prototypes PGCD/LD rely on (Dice ↓). Zero-shot wins: +0.035 Dice, zero GPU-hours. This also means the whole framework can be deployed on a new organ or scanner protocol without ever touching MedSAM itself, only the lightweight student is trained.

Conclusion & future work

A frozen, zero-shot MedSAM improves compact, prompt-free students when transfer goes **beyond pixel-wise response matching**. Structured transfer, not raw teacher accuracy, drives the gains; ground-truth supervision stays dominant, so the student does not inherit teacher errors.

Future work. Multi-organ and multi-modality (CT + MRI); volumetric (3D) topology consistency; box-free teacher prompting.

In short: TopoMedKD transfers MedSAM knowledge into a compact, prompt-free model suitable for efficient automatic segmentation.

Acknowledgments

Supported by the Spanish Ministry: PID2023-146925OB-I00.

References

- [1] Ma et al. *MedSAM*, 2024. [2] Kirillov et al. *Segment Anything*, 2023. [3] Heller et al. *KiTS19*, 2021.

Paper

